Why Multilevel Models Are Good Models For Longitudinal Data

Multilevel Models Offer An Incredibly Flexible Treatment of Time and Time Varying Processes and Covariates

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1 Key Idea

> Multilevel Models are Useful for Complicated Longitudinal Data and Research Designs
>
> Multilevel models are uniquely suited for complicated longitudinal data and research designs. Among other advantages, multilevel models are able to easily accommodate designs where individuals are observed at different times and different numbers of times.[1]

2 Visually

3 Data Structures

Multilevel models for longitudinal data prefer data in long format.

[1] Many, if not most, of the advantages listed in this handout will also apply to any approach using long data.

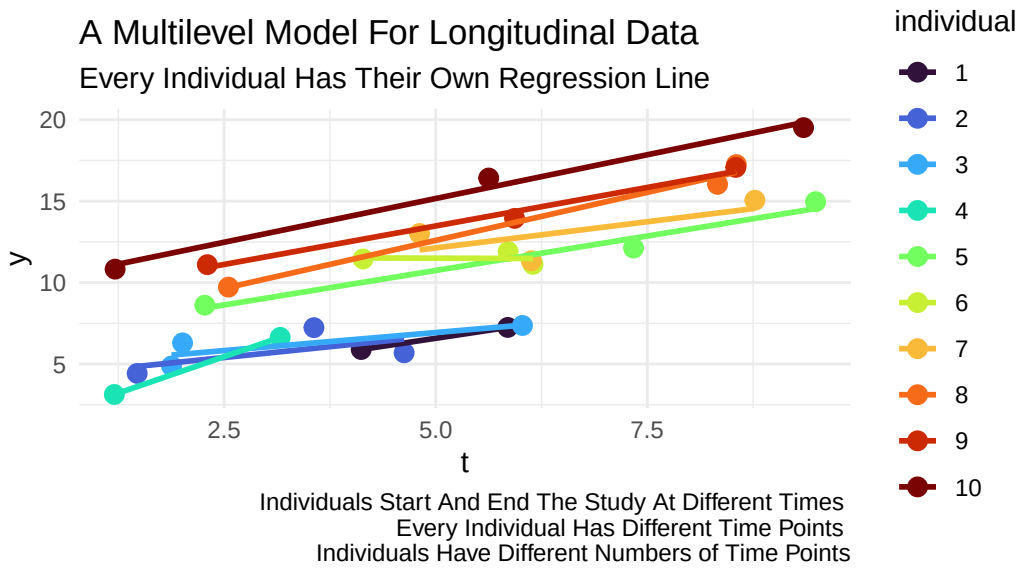


Figure 1: A Multilevel Model For Longitudinal Data

Table 1: Data in WIDE format

id	x1	x2	x3	y1	y2	y3
1						
2						
3						

Table 2: Data in LONG format

id	t	x	y
1	1		
1	2		
1	3		
2	1		
2	2		
2	3		
3	1		
3	2		
3	3		

4 Equation

$$y_{it} = \beta_0 + \beta_1 t_{it} + \beta_2 x_{it} + u_{0i} + e_{it} \tag{1}$$

Person-Observations

Every row is a *person-observation* (person i observed at time t). Every person has *multiple rows*.

5 Advantages of the Multilevel Model for Longitudinal Data

1. Using the algebra in Equation 1, these models can easily accommodate both time varying and time invariant variables (Hox, 2010; Hox et al., 2018; Rabe-Hesketh & Skrondal, 2022; Raudenbush & Bryk, 2002; Singer & Willett, 2003).
2. There is no multicollinearity issue with multiple β coefficients for multiple waves of data. By inspection of Equation 1, we see that there is only a single β coefficient for each variable, $\therefore$ no multicollinearity problem.
3. *Unbalanced data is less of a problem*, the data structure and estimation are robust to these possibilities (Raudenbush, 1995; Raudenbush & Bryk, 2002; Singer & Willett, 2003).
4. *Missing data is less of a problem* (assuming *MCAR*). When a person observation is missing, that person simply has fewer rows of data (Hox, 2010; Luke, 2004; Rabe-Hesketh & Skrondal, 2022; Raudenbush, 1995; Raudenbush & Bryk, 2002). But all rows of data are "matched" to the same person by i.
5. We have an *explicit function of time* $\beta_1 t$, and could treat time more flexibly, by creating a polynomial function of time e.g. by adding $\beta_2 t^2$, etc. (Raudenbush & Bryk, 2002; Singer & Willett, 2003). (We could even substitute $\beta \ln(t)$.)
6. Again, by inspection of Equation 1, we see that *multiple or many time-points are not a problem*. We would use the same algebra for 2 time points or for 10,000 time points. (Helpful when we start to think about intensive longitudinal data *e.g.* George Holden's *recording study*).
7. We are *measuring exactly the time at which events take place* for each individual (Luke, 2004; Singer & Willett, 2003). Not simply saying *Wave 1*, *Wave 2*, *Wave 3*, etc...
8. Unequally spaced time points are not a problem (Raudenbush, 1995). Every individual could have a *completely different set of time points* and even a *completely different number of time points* (Hox, 2010; Hox et al., 2018; Luke, 2004; Singer & Willett, 2003).

Key Idea

Put more intuitively, (3) & (4) mean that individuals can enter the study late, can leave the study early, or can miss assessment points in the middle of the study without major problems for estimation of the model. (8) means that individual can be assessed at very different times, or even different numbers of times without presenting major issues for estimation.

The Possibility of Selection Bias With Missing Data

We do need to think very carefully about *why* individuals are missing from the data. Are they missing effectively at random? Or are there structural reasons that individuals may not be part of certain study waves? Often the individuals who have experienced the most trauma, violence or discrimination, or who have the lowest incomes are the ones who are not able to participate in every survey wave. While multilevel models are robust to these situations,

we need to think carefully about whether we need to account for this possible selection bias, perhaps by imputing missing data.

What Is The Appropriate Metric For Time?

We do need to think carefully about what is the appropriate variable for time. Is it the variable we used to reshape the data–often `wave`–or some other more appropriate metric, like `age` or `time in study` (Singer & Willett, 2003)?

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